

Chapter 14

Applications to Group Theory

In the preceding chapter, we showed how a problem of topology—classifying all covering spaces of a space B —can be reduced to a problem of algebra—classifying all subgroups of the fundamental group of B . Now we consider the reverse process, that of reducing a problem of algebra to one of topology. The problem of algebra in question is that of showing that any subgroup of a free group is itself a free group. While this statement is certainly believable, it is not one whose proof is obvious. We shall proceed by applying the theory of covering spaces to certain topological spaces called *linear graphs*.

§83 Covering Spaces of a Graph

We define here the notion of linear graph (introduced earlier in the finite case), and prove the basic theorem that any covering space of a linear graph is itself a linear graph.

Recall that an *arc* A is a space homeomorphic to the unit interval $[0, 1]$. The *end points* of A are the points p, q corresponding to 0 and 1 under the homeomorphism; they are the unique points of A such that $A - p$ and $A - q$ are connected. The *interior* of an arc A consists of A with its end points deleted.

Definition. A *linear graph* is a space X that is written as the union of a collection of subspaces A_α , each of which is an arc, such that:

- (1) The intersection $A_\alpha \cap A_\beta$ of two arcs is either empty or consists of a single point that is an end point of each.
- (2) The topology of X is coherent with the subspaces A_α .

The arcs A_α are called the *edges* of X , and their interiors are called the *open edges* of X . Their end points are called the *vertices* of X ; we denote the set of vertices of X by X^0 .

If X is a linear graph, and if C is a subset of X that equals a union of edges and vertices of X , then C is closed in X . For the intersection of C with A_α is closed in A_α , since it is either empty, or it equals A_α , or it equals one or both vertices of A_α . It follows that each edge of X is a closed subset of X . It also follows that X^0 is a closed discrete subspace of X , since any subset of X^0 is closed in X .

In the case of a finite graph, considered earlier, we used the Hausdorff condition in our definition in place of condition (2), it followed, in that case, that the topology of X was coherent with the subspaces A_α . In the case of an infinite graph, this would no longer be true, so we must assume the coherence condition as part of the definition. We would assume the Hausdorff condition as well, but it is no longer necessary, for it follows from the coherence condition:

Lemma 83.1. *Every linear graph X is Hausdorff; in fact, it is normal.*

Proof. Let B and C be disjoint closed subsets of X . Assume, without loss of generality, that every vertex of X belongs either to B or to C . For each α , choose disjoint subsets U_α and V_α of A_α that are open in A_α , containing $B \cap A_\alpha$ and $C \cap A_\alpha$, respectively. Let $U = \bigcup U_\alpha$ and $V = \bigcup V_\alpha$. Then U and V contain B and C , respectively.

We show the sets U and V are disjoint. If $x \in U \cap V$, then $x \in U_\alpha \cap V_\beta$ for some $\alpha \neq \beta$. This fact implies that A_α and A_β contain the point x , which means that x is a vertex of X . This is impossible, for if $x \in B$, then x lies in no set V_β , and if $x \in C$, then x lies in no set U_α .

Now we show U and V are open in X . To show U is open, we show that $U \cap A_\alpha = U_\alpha$ for each α . By definition, $U \cap A_\alpha$ contains U_α . If x is a point of $U \cap A_\alpha$ not in U_α , then x belongs to U_β for some $\beta \neq \alpha$. Then both A_β and A_α contain x , so that x must be a vertex of X . This is impossible, for if $x \in B$, then $x \in U_\alpha$ by definition of U_α , and if $x \in C$, then x cannot belong to U . ■

EXAMPLE 1. If X is the wedge of the circles S_α , with common point p , then X can be expressed as a linear graph. We need merely write each S_α as a graph having three edges, with p as one of its vertices; then X is a union of arcs. To show that the topology of the wedge X is coherent with the resulting collection of arcs, we note that if $D \cap A_\alpha$ is closed in A_α for each arc A_α , then $D \cap S_\beta$ is the union of three sets of the form $D \cap A_\alpha$ and so is closed in S_β ; then D is closed in X by definition. See Figure 83.1.

EXAMPLE 2. Let J be a discrete space, and let $E = [0, 1] \times J$. Then the quotient space X obtained from E by collapsing the set $\{0\} \times J$ to a point p is a linear graph.

The quotient map $\pi : E \rightarrow X$ is a closed map. For if C is closed in E , then $\pi^{-1}\pi(C)$ equals $C \cup (\{0\} \times J)$ if C contains a point of $\{0\} \times J$, and $\pi^{-1}\pi(C)$ equals C otherwise. In either case, $\pi^{-1}\pi(C)$ is closed in E , so that $\pi(C)$ is closed in X . It follows that π maps each space $[0, 1] \times \alpha$ homeomorphically onto its image A_α , so that A_α is an arc. The topology of X is coherent with the subspaces A_α because π is a quotient map. See Figure 83.2.

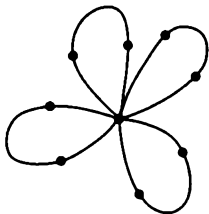


Figure 83.1

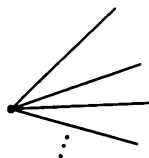


Figure 83.2

Definition. Let X be a linear graph. Let Y be a subspace of X that is a union of edges of X . Then Y is closed in X and is itself a linear graph; we call it a **subgraph** of X .

To show that Y is a linear graph, we need to show that the subspace topology on Y is coherent with the set of edges of Y . If the subset D of Y is closed in the subspace topology, then D is closed in X , so that $D \cap A_\alpha$ is closed in A_α for each edge of X , and in particular for each edge of Y . Conversely, suppose $D \cap A_\beta$ is closed in A_β for each edge A_β of Y . We must show that $D \cap A_\alpha$ is closed in A_α for each edge A_α of X that is *not* contained in Y . But in this case, $D \cap A_\alpha$ is either empty or a one-point set! We conclude that Y has the topology coherent with its set of edges.

Lemma 83.2. Let X be a linear graph. If C is a compact subspace of X , there exists a finite subgraph Y of X that contains C . If C is connected, Y can be chosen to be connected.

Proof. First, note that C contains only finitely many vertices of X . For $C \cap X^0$ is a closed discrete subspace of the compact space C ; since it has no limit point, it must be finite. Similarly, there are only finitely many values of α for which C contains an interior point of the edge A_α . For if we choose a point x_α of C interior to A_α for each index α for which it is possible to do so, we obtain a collection $B = \{x_\alpha\}$ whose intersection with each edge A_β is a one-point set or empty. It follows that every subset of B is closed in X , so that B is a closed discrete subspace of C and hence finite.

Form Y by choosing, for each vertex x of X belonging to C , an edge of X having x as a vertex, and adjoining to these edges all edges A_α whose interiors contain points

of C . Then Y is a finite subgraph containing C . Note that if C is connected, then Y is the union of a collection of arcs each of which intersects C , so that Y is connected. ■

Lemma 83.3. *If X is a linear graph, then X is locally path connected and semilocally simply connected.*

Proof. *Step 1.* We show X is locally path connected. If $x \in X$ and x lies interior to some edge of X , then within every neighborhood of x is a neighborhood of x homeomorphic to an open interval of $\mathbb{R}$, which is path connected. On the other hand, if x is a vertex of X and U is a neighborhood of x , then we can choose, for each edge A_α having x as an end point, a neighborhood V_α of x in A_α lying in U that is homeomorphic to the half-open interval $[0, 1)$. Then $\bigcup V_\alpha$ is a neighborhood of x in X lying in U , and it is a union of path-connected spaces having the point x in common.

Step 2. We show X is semilocally simply connected. Indeed, we show that if $x \in X$, then x has a neighborhood U such that $\pi_1(U, x)$ is trivial.

If x lies interior to some edge of X , then the interior of this edge is such a neighborhood. So suppose x is a vertex of X . Let $\overline{St}x$ denote the union of those edges of X that have x as an end point, and let Stx denote the subspace of $\overline{St}x$ obtained by deleting all vertices other than x . (Stx is called the *star* of x in X .) The set Stx is open in X , since its complement is a union of arcs and vertices. We show that $\pi_1(Stx, x)$ is trivial.

Let f be a loop in Stx based at x . Then the image set $f(I)$ is compact, so it lies in some finite union of arcs of $\overline{St}x$. Any such union is homeomorphic to the union of a finite set of line segments in the plane having an end point in common. And for any loop in such a space, the straight-line homotopy will shrink it to the constant loop at x . ■

Now if x is a vertex of X , it is in fact true that the one-point space $\{x\}$ is a deformation retract of $\overline{St}x$. But there is a surprising amount of effort required to show that the obvious deformation is continuous. One needs the fact that a map

$$F : (\overline{St}x) \times I \rightarrow \overline{St}x$$

is continuous if its restriction to each subspace $A_\alpha \times I$ is continuous. This result follows from the pasting lemma in the case where $\overline{St}x$ is a union of only finitely many arcs, but the general result requires one to show that the topology of $(\overline{St}x) \times I$ is coherent with the subspaces $A_\alpha \times I$. This in turn follows from a basic theorem about products of quotient maps. (See Exercise 11 of §29.) These considerations do not arise if one wishes merely to shrink a loop to a point (rather than shrinking the entire space $\overline{St}x$), since any loop lies in the union of a finite number of edges, where there is no problem.

Now we discuss covering spaces of linear graphs. Note that the convention that every covering space is assumed to be path connected and locally path connected, which we assumed in the last chapter, no longer applies.

Theorem 83.4. Let $p : E \rightarrow X$ be a covering map, where X is a linear graph. If A_α is an edge of X and B is a path component of $p^{-1}(A_\alpha)$, then p maps B homeomorphically onto A_α . Furthermore, the space E is a linear graph, with the path components of the spaces $p^{-1}(A_\alpha)$ as its edges.

Proof. *Step 1.* We show that p maps B homeomorphically onto A_α . Because the arc A_α is path connected and locally path connected, Theorems 53.2 and 80.1 tell us that the map $p_0 : B \rightarrow A_\alpha$ obtained by restricting p is a covering map. Because B is path connected, the lifting correspondence $\phi : \pi_1(A_\alpha, a) \rightarrow p_0^{-1}(a)$ is surjective; because A_α is simply connected, $p_0^{-1}(a)$ consists of a single point. (See Theorem 54.4.) Hence p_0 is a homeomorphism.

Step 2. Because X is the union of the arcs A_α , the space E is the union of the arcs B that are the path components of the spaces $p^{-1}(A_\alpha)$. Let B and B' be path components of $p^{-1}(A_\alpha)$ and $p^{-1}(A_\beta)$, respectively, with $B \neq B'$. We show B and B' intersect in at most a common end point. If A_α and A_β are equal, then B and B' are disjoint, and if A_α and A_β are disjoint, so are B and B' . Therefore, if B and B' intersect, A_α and A_β must intersect in an end point x of each; then $B \cap B'$ consists of a single point, which must be an end point of each.

Step 3. We show that E has the topology coherent with the arcs B . This is the hardest part of the proof. Let W be a subset of E such that $W \cap B$ is open in B , for each arc B of E . We show that W is open in E .

First, we show that $p(W)$ is open in X . If A_α is an edge of X , then $p(W) \cap A_\alpha$ is the union of the sets $p(W \cap B)$, as B ranges over all path components of $p^{-1}(A_\alpha)$. Each of these sets $p(W \cap B)$ is open in A_α , because p maps B homeomorphically onto A_α ; hence their union $p(W) \cap A_\alpha$ is open in A_α . Because X has the topology coherent with the subspaces A_α , the set $p(W)$ is open in X .

Second, we prove our result in the special case where the set W is contained in one of the slices V of $p^{-1}(U)$, where U is an open set of X that is evenly covered by p . By the result just proved, we know that the set $p(W)$ is open in X . It follows that $p(W)$ is open in U . Because the map of V onto U obtained by restricting p is a homeomorphism, W must be open in V , and hence open in E .

Finally, we prove our result in general. Choose a covering $\mathcal{A}$ of X by open sets U that are evenly covered by p . Then the slices V of the sets $p^{-1}(U)$, for $U \in \mathcal{A}$, cover E . For each such slice V , let $W_V = W \cap V$. The set W_V has the property that for each arc B of E , the set $W_V \cap B$ is open in B , for $W_V \cap B = (W \cap B) \cap (V \cap B)$ and both $W \cap B$ and $V \cap B$ are open in B . The result of the preceding paragraph implies that W_V is open in E . Since W is the union of the sets W_V , it also is open in E . ■

Exercises

1. In the proof of normality of a linear graph X , why did we assume that every vertex of X belongs either to B or to C ?

2. The *Euler number* of a finite linear graph X equals the number of vertices of X minus the number of edges of X . It is in fact a topological invariant of X , as we shall see. What is the Euler number of an arc? a circle? a wedge of n circles? the complete graph on n vertices? If E is an n -fold covering space of X , how are the Euler numbers of E and X related?

§84 The Fundamental Group of a Graph

Now we prove the basic theorem that the fundamental group of any linear graph is a free group. Henceforth we shall refer to a linear graph simply as a *graph*.

Definition. An *oriented edge* e of a graph X is an edge of X together with an ordering of its vertices; the first is called the *initial vertex*, and the second, the *final vertex*, of e . An *edge path* in X is a sequence $e_1, \dots, e_n$ of oriented edges of X such that the final vertex of e_i equals the initial vertex of e_{i+1} , for $i = 1, \dots, n-1$. Such an edge path is entirely specified by the sequence of vertices $x_0, \dots, x_n$, where x_0 is the initial vertex of e_1 and x_i is the final vertex of e_i for $i = 1, \dots, n$. It is said to be an edge path *from* x_0 *to* x_n . It is called a *closed edge path* if $x_0 = x_n$.

Given an oriented edge e of X , let f_e be the positive linear map of $[0, 1]$ onto e ; it is a path from the initial point of e to the final point of e . Then, corresponding to the edge path $e_1, \dots, e_n$ from x_0 to x_n , one has the actual path

$$f = f_1 * (f_2 * (\dots * f_n))$$

from x_0 to x_n , where $f_i = f_{e_i}$; it is uniquely determined by the edge path $e_1, \dots, e_n$. We call it the *path corresponding to the edge path* $e_1, \dots, e_n$. If the edge path is closed, then the corresponding path f is a loop.

Lemma 84.1. A graph X is connected if and only if every pair of vertices of X can be joined by an edge path in X .

Proof. Suppose X is connected. Define $x \sim y$ if there is an edge path in X from x to y . For any edge of X , its end points belong to the same equivalence class; let Y_x denote the union of all edges whose end points are equivalent to x . Then Y_x is a subgraph of X and hence is closed in X . The subgraphs Y_x form a partition of X into disjoint closed subspaces; since X is connected, there must be only one such.

Conversely, suppose every pair of vertices of X can be joined by an edge path. Then they can be joined by an actual path in X . Hence all the vertices of X belong to the same component of X . Since each edge is connected, it also belongs to this component. Thus X is connected. ■

Definition. Let $e_1, \dots, e_n$ be an edge path in the linear graph X . It can happen that for some i , the oriented edges e_i and e_{i+1} consist of the same edge of X , but with opposite orientations. If this situation does *not* occur, then the edge path is said to be a **reduced edge path**.

Note that if this situation *does* occur, then one can delete e_i and e_{i+1} from the sequence of oriented edges and still have an edge path remaining (provided the original sequence consists of at least three edges). This deletion process is called **reducing** the edge path. It enables one to show that in any connected graph, every pair of distinct vertices can be joined by a reduced edge path. See Figure 84.1.

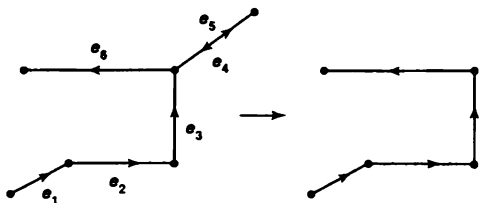


Figure 84.1

Definition. A subgraph T of X is said to be a **tree** in X if T is connected and T contains no closed reduced edge paths.

A linear graph consisting of a single edge is a tree. The graph in Figure 84.2 is not a tree, but deletion of the edge e would make it a tree. The graph in Figure 84.3 is a tree; deletion of the edge A would leave a tree remaining.

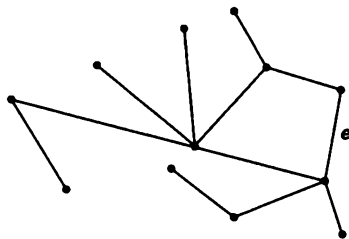


Figure 84.2

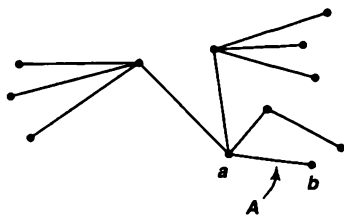


Figure 84.3

Lemma 84.2. If T is a tree in X , and if A is an edge of X that intersects T in a single vertex, then $T \cup A$ is a tree in X . Conversely, if T is a finite tree in X that consists of more than one edge, then there is a tree T_0 in X and an edge A of X that intersects T_0 in a single vertex, such that $T = T_0 \cup A$.

Proof. Suppose T is a tree in X and A is an edge that intersects T in a single vertex. Clearly $T \cup A$ is connected; we show it contains no closed reduced edge paths. Let a and b be the end points of A , with $\{a\} = T \cap A$. See Figure 84.3. Suppose $x_0, \dots, x_n = x_0$ is the vertex sequence of a closed reduced edge path in $T \cup A$. If none of the vertices x_i equals b , then the edge path lies in T , contrary to hypothesis. If $x_i = b$ for some i with $0 < i < n$, then we must have $x_{i-1} = a$ and $x_{i+1} = a$; hence the edge path is not reduced, contrary to hypothesis. Finally, if $x_0 = b = x_n$ and $x_i \neq b$ for $i = 1, \dots, n-1$, then $x_1 = a$ and $x_{n-1} = a$, and the vertex sequence $x_1, \dots, x_{n-1}$ specifies a closed reduced edge path in T , again contrary to hypothesis.

Now let T be a finite tree in X having more than one edge. First, we show that some vertex b of T belongs to only one edge of T . If this is not the case, we can construct an edge path in T as follows: Begin with a vertex x_0 of T ; then choose an edge e_1 of T having x_0 as an end point. Orient e_1 so x_0 is its initial vertex. Let x_1 be the other end point of e_1 , and let e_2 be an edge of T different from e_1 having x_1 as a vertex. Orient e_2 so x_1 is its initial vertex. Similarly continue. No two successive terms of the sequence $e_1, e_2, \dots$ are opposite orientations of the same edge of T . Since T is finite, there must be an index n such that $x_n = x_i$ for some $i < n$. Then the sequence of vertices $x_i, x_{i+1}, \dots, x_n$ determines a closed reduced edge path in T , contrary to hypothesis. See Figure 84.4.

Let b be a vertex of T belonging to only one edge A of T , and let T_0 consist of all edges of T different from A . Then $T = T_0 \cup A$. Because T is connected, T_0 must intersect A in its other vertex a . We show T_0 is a tree. Clearly T_0 contains no closed reduced edge paths, because T contains none. Furthermore, T_0 is connected. For if T_0 were the union of two disjoint closed sets C and D , the point a would lie in one of them, say C . Then $C \cup A$ and D would be disjoint closed sets whose union is T , contrary to the fact that T is connected. ■

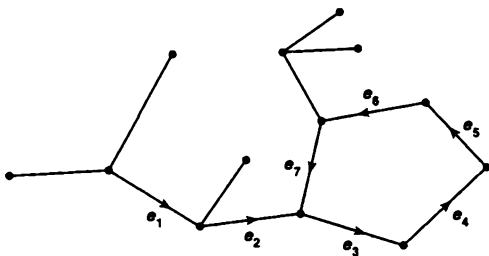


Figure 84.4

Theorem 84.3. Any tree T is simply connected.

Proof. We first consider the case where T is a finite tree. If T consists of a single edge, then T is simply connected. If T has n edges with $n > 1$, there is an edge A

of T such that $T = T_0 \cup A$, where T_0 is a tree with $n - 1$ edges and $T_0 \cap A$ is a single vertex. Then T_0 is a deformation retract of T . Since T_0 is simply connected by the induction hypothesis, so is T .

To prove the general case, let f be a loop in T . The image set of f is compact and connected, so it is contained in a finite connected subgraph Y of T . Now Y contains no closed reduced edge paths, because T contains none. Thus Y is a tree. Since Y is finite, it is simply connected. Hence f is path homotopic to a constant in Y . ■

Definition. A tree T in X is *maximal* if there is no tree in X that properly contains T .

Theorem 84.4. Let X be a connected graph. A tree T in X is maximal if and only if it contains all the vertices of X .

Proof. Suppose T is a tree in X that contains all the vertices of X . If Y is a subgraph of X that properly contains T , we show that Y contains a closed reduced edge path; it follows that T is maximal. Let A be an edge of Y that is not in T ; by hypothesis, the end points a and b of A belong to T . Since T is connected, we can choose a reduced edge path $e_1, \dots, e_n$ in T from a to b . If we follow this sequence by the edge A , oriented from b to a , we obtain closed reduced edge path in Y .

Now let T be a tree in X that does not contain all the vertices of X . We show T is not maximal. Let x_0 be a vertex of X not in T . Since X is connected, we may choose an edge path in X from x_0 to a vertex of T , specified by the sequence of vertices $x_0, \dots, x_n$. Let i be the smallest index such that $x_i \in T$. Let A be the edge of X with vertices x_{i-1} and x_i . Then $T \cup A$ is a tree in X , by the preceding lemma, and $T \cup A$ properly contains T . ■

Theorem 84.5. If X is a linear graph, every tree T_0 in X is contained in a maximal tree in X .

Proof. We apply Zorn's lemma to the collection $\mathcal{T}$ of all trees in X that contain T_0 , strictly partially ordered by proper inclusion. To show this collection has a maximal element, we need only prove the following:

If $\mathcal{T}'$ is a subcollection of $\mathcal{T}$ that is simply ordered by proper inclusion, then the union Y of the elements of $\mathcal{T}'$ is a tree in X .

To begin, we note that since Y is a union of subgraphs of X , it is a subgraph of X . Second, since Y is a union of connected spaces that contain the connected space T_0 , the space Y is connected.

Finally, we suppose that $e_1, \dots, e_n$ is a closed reduced edge path in Y and derive a contradiction. For each i , choose an element T_i of $\mathcal{T}'$ that contains e_i . Because $\mathcal{T}'$ is simply ordered by proper inclusion, one of the trees $T_1, \dots, T_n$, say T_j , contains all the others. But then $e_1, \dots, e_n$ is a closed reduced edge path in T_j , contrary to hypothesis. ■

Now we compute the fundamental group of a graph. We need the following result.

Lemma 84.6. Suppose $X = U \cup V$, where U and V are open sets of X . Suppose that $U \cap V$ is the union of two disjoint open path-connected sets A and B , that α is a path in U from the point a of A to the point b of B , and that β is a path in V from b to a . If U and V are simply connected, then the class $[\alpha * \beta]$ generates $\pi_1(X, a)$.

Proof. The situation is similar to that of Theorem 59.1, except that $U \cap V$ has two path components instead of one. The proof is also similar.

Let f be a loop in X based at a . Choose a subdivision $0 = a_0 < a_1 < \cdots < a_n = 1$ of $[0, 1]$ such that for each i , $f(a_i) \in U \cap V$ and f maps $[a_{i-1}, a_i]$ into either U or V . Let f_i be the positive linear map of $[0, 1]$ onto $[a_{i-1}, a_i]$ followed by f ; then $[f] = [f_1] * \cdots * [f_n]$. For $i = 1, \dots, n-1$, choose a path α_i in either A or B from a or b to $f(a_i)$; choose α_0 and α_n to be the constant paths at a . Then set

$$g_i = \alpha_{i-1} * (f_i * \bar{\alpha}_i).$$

By direct computation, $[f] = [g_1] * \cdots * [g_n]$. Because g_i is a path in U or in V with end points in the set $\{a, b\}$, and because U and V are simply connected, g_i is path homotopic either to a constant or to α , β , $\bar{\alpha}$, or $\bar{\beta}$. It follows that either $[f]$ is trivial, or it equals a positive power of $[\alpha * \beta]$ or $[\bar{\beta} * \bar{\alpha}]$. Hence $[\alpha * \beta]$ generates the group $\pi_1(X, a)$. See Figure 84.5. ■

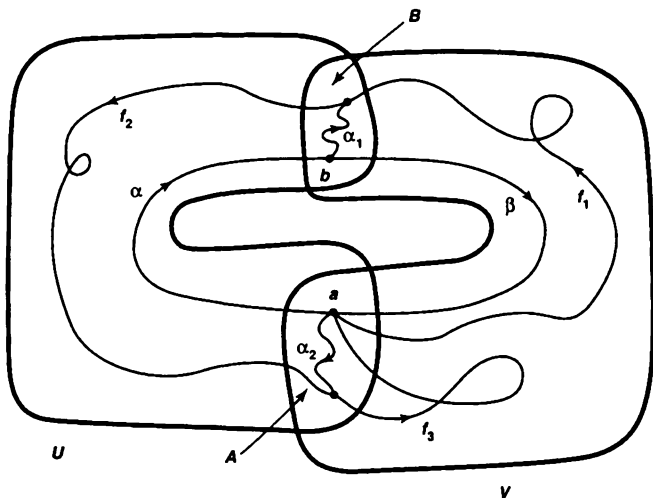


Figure 84.5

Theorem 84.7. *Let X be a connected graph that is not a tree. Then the fundamental group of X is a nontrivial free group. Indeed, if T is a maximal tree in X , then the fundamental group of X has a system of free generators that is in bijective correspondence with the collection of edges of X that are not in T .*

Proof. Let T be a maximal tree in X ; it contains all the vertices of X . Let x_0 be a fixed vertex of T . For each vertex x of X , choose a path γ_x in T from x_0 to x . Then for each edge A of X that is not in T , define a loop g_A in X as follows. Orient A ; let f_A be the linear path in A from its initial end point x to its final end point y ; and set

$$g_A = \gamma_x * (f_A * \bar{\gamma}_y).$$

We show that the classes $[g_A]$ form a system of free generators for $\pi_1(X, x_0)$.

Step 1. We first prove the theorem when the edges of X not in T are finite in number. We proceed by induction. The induction step is easy, so we consider it first.

Let $A_1, \dots, A_n$ be the edges of X not in T , where $n > 1$. Orient these edges and let g_i denote the loop g_{A_i} . For each i , choose a point p_i interior to A_i . Let

$$U = X - p_2 - \dots - p_n \quad \text{and} \quad V = X - p_1.$$

Then U and V are open in X , and the space $U \cap V = X - p_1 - \dots - p_n$ is simply connected, since it has T as a deformation retract. Therefore, $\pi_1(X, x_0)$ is the free product of the groups $\pi_1(U, x_0)$ and $\pi_1(V, x_0)$, by Corollary 70.3.

The space U has $T \cup A_1$ as a deformation retract, so $\pi_1(U, x_0)$ is free on the generator $[g_1]$, as we shall prove in Step 2. The space V has $T \cup A_2 \cup \dots \cup A_n$ as a deformation retract, so it is free on the generators $[g_2], \dots, [g_n]$, by the induction hypothesis. It follows from Theorem 69.2 that $\pi_1(X, x_0)$ is free on the generators $[g_1], \dots, [g_n]$. See Figure 84.6.

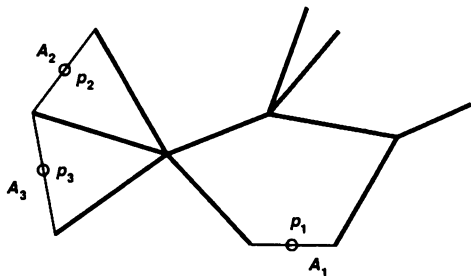


Figure 84.6

Step 2. We now consider the case where there is only one edge D of X that is not in T . This step is more difficult. Orient D . We show $\pi_1(X, x_0)$ is infinite cyclic with generator $[g_D]$.

Let a_0 and a_1 be the initial and final points of D , respectively. Let us write D as the union of three arcs: D_1 , with end points a_0 and a , D_2 , with end points a and b , and D_3 , with end points b and a_1 . See Figure 84.7. Let f_1 , f_2 , and f_3 be the linear paths in D from a_0 to a , and a to b , and b to a_1 , respectively. We apply the preceding theorem to compute $\pi_1(X, a)$.

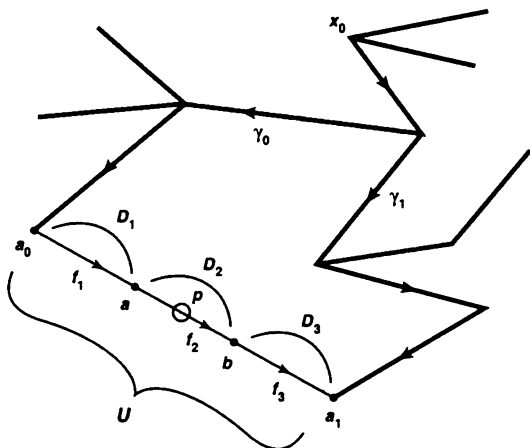


Figure 84.7

Choose a point p interior to the arc D_2 . Set $U = D - a_0 - a_1$ and $V = X - p$. Then U and V are open sets in X whose union is X . The space U is simply connected because it is an open arc. And the space V is simply connected because it has the tree T as a deformation retract. The space $U \cap V$ equals $U - p$; it has two path components; let A be the one containing a and let B be the one containing b . Then the hypotheses of the preceding lemma are satisfied. The path $\alpha = f_2$ is a path in U from a to b . If we set $\gamma_0 = \gamma_{a_0}$ and $\gamma_1 = \gamma_{a_1}$, then the path $\beta = (f_3 * (\bar{\gamma}_1 * (\gamma_0 * f_1)))$ is a path in V from b to a . Therefore, $\pi_1(X, a)$ is generated by the class

$$[\alpha * \beta] = [f_2] * [f_3] * [\bar{\gamma}_1] * [\gamma_0] * [f_1].$$

It follows that $\pi_1(X, x_0)$ is generated by $\hat{\delta}[\alpha * \beta]$, where $\hat{\delta}$ is the path $\bar{f}_1 * \bar{\gamma}_0$ from a to x_0 . We compute this path-homotopy class as follows:

$$\begin{aligned} \hat{\delta}[\alpha * \beta] &= [\gamma_0 * f_1] * [\alpha * \beta] * [\bar{f}_1 * \bar{\gamma}_0] \\ &= [\gamma_0] * [f_1 * (f_2 * f_3)] * [\bar{\gamma}_1] \\ &= [\gamma_0] * [f_D] * [\bar{\gamma}_1] \\ &= [g_D]. \end{aligned}$$

Therefore, $[g_D]$ generates $\pi_1(X, x_0)$.

It remains to show that the element $[g_D]$ has infinite order, so that $\pi_1(X, x_0)$ is infinite cyclic. One can apply Theorem 63.1 (which we used in proving the Jordan curve theorem), which states that $[\alpha * \beta]$ has infinite order in $\pi_1(X, a)$. Alternatively (and more easily), one can consider the map $\pi : X \rightarrow S^1$ that collapses the tree T to a single point p and maps the open arc $\text{Int } D$ homeomorphically onto $S^1 - p$. Then $\pi \circ \gamma_0$ and $\pi \circ \bar{\gamma}_1$ are constant paths, so that

$$\pi_*([g_D]) = [\pi \circ f_D].$$

This class generates $\pi_1(S^1, p)$. It follows that $[g_D]$ has infinite order in $\pi_1(X, x_0)$.

Step 3. Now we consider the situation where the collection of edges of X not in T is infinite. The proof in this case is so similar to the corresponding proof for an infinite wedge of circles that we omit the details. (See Theorem 71.3.) The crucial facts are these: Any loop in X based at x_0 lies in the space

$$X(\alpha_1, \dots, \alpha_n) = T \cup A_{\alpha_1} \cup \dots \cup A_{\alpha_n}$$

for some finite set of indices α_i , and any path homotopy between such loops also lies in such a space. By this means the general case is reduced to the finite case. ■

Exercises

1. Give an example to show that the second part of Lemma 84.2 need not hold if T is infinite.
2. What is the cardinality of a system of free generators for the fundamental group of the complete graph on n vertices? of the utilities graph? (See §64.)
3. Let X be the wedge of two circles; let $p : E \rightarrow X$ be a covering map. The fundamental group of E maps isomorphically under p_* onto a subgroup H of the fundamental group of X ; the latter is free on two generators α and β .
 - (a) For each of the four covering spaces E given in Exercise 2 of §81, determine the cardinality of a system of free generators for the fundamental group of E .
 - (b) For each of these covering spaces, find, in terms of α and β , a system of free generators for the subgroup H of the fundamental group of X .

§85 Subgroups of Free Groups

We now prove our main theorem, to the effect that a subgroup H of a free group F is free. The method of proof, remarkably enough, will give us some information about the cardinality of a system of free generators for H , when the cardinality of a system of free generators for F is known.

Theorem 85.1. *If H is a subgroup of a free group F , then H is free.*

Proof. Let $\{\alpha \mid \alpha \in J\}$ be a system of free generators for F . Let X be a wedge of circles S_α , one for each $\alpha \in J$; let x_0 be their common point. We can give X the structure of a linear graph by breaking each circle S_α into three arcs, two of which have x_0 as an end point. The function that assigns to each α , a loop generating $\pi_1(S_\alpha, x_0)$, induces an isomorphism of F with $\pi_1(X, x_0)$. Therefore we may as well assume that F equals the group $\pi_1(X, x_0)$.

The space X is path connected, locally path connected, and semilocally simply connected. Therefore Theorem 82.1 applies to show that there exists a path-connected covering space $p: E \rightarrow X$ of X such that, for some point e_0 of $p^{-1}(x_0)$,

$$p_*(\pi_1(E, e_0)) = H.$$

Since p_* is a monomorphism, $\pi_1(E, e_0)$ is isomorphic to H .

The space E is a linear graph, by Theorem 83.4. Then Theorem 84.7 implies that its fundamental group is a free group. ■

Definition. If X is a finite linear graph, we define the **Euler number** of X to be the number of vertices of X minus the number of edges. It is commonly denoted by the Greek letter chi, as $\chi(X)$.

Lemma 85.2. *If X is a finite, connected linear graph, then the cardinality of a system of free generators for the fundamental group of X is $1 - \chi(X)$.*

Proof. *Step 1.* We first show that for any finite tree T , we have $\chi(T) = 1$. We proceed by induction on the number n of edges in T . If $n = 1$, then T has one edge and two vertices, so $\chi(T) = 1$. If $n > 1$, we can write $T = T_0 \cup A$, where T_0 is a tree having $n - 1$ edges, and A is an edge that intersects T_0 in a single vertex. We have $\chi(T_0) = 1$ by the induction hypothesis. The graph T has one more edge and one more vertex than T_0 ; hence $\chi(T) = \chi(T_0)$.

Step 2. We prove the theorem. Given X , let T be a maximal tree in X . If $X = T$, we are finished. Otherwise, let $A_1, \dots, A_n$ be the edges of X that are not in T . Then the fundamental group of X has a system of n free generators. On the other hand, X and T have exactly the same vertex set, and X has n more edges than T . Hence,

$$\chi(X) = \chi(T) - n = 1 - n,$$

so that $n = 1 - \chi(X)$. ■

Definition. Let H be a subgroup of the group G . If the collection G/H of right cosets of H in G is finite, its cardinality is called the **index** of H in G . (The collection of left cosets of H in G has the same cardinality, of course.)

Theorem 85.3. Let F be a free group with $n + 1$ free generators; let H be a subgroup of F . If H has index k in F , then H has $kn + 1$ free generators.

Proof. We apply the construction given in the proof of Theorem 85.1. We can assume that $F = \pi_1(X, x_0)$, where X is a linear graph whose underlying space is a wedge of $n + 1$ circles. Given H , we choose a path-connected covering space $p : E \rightarrow X$ such that $p_*(\pi_1(E, e_0)) = H$. Now the lifting correspondence

$$\Phi : \pi_1(X, x_0)/H \rightarrow p^{-1}(x_0)$$

is a bijection. Therefore, E is a k -fold covering of X .

The space E is also a linear graph. Given an edge A of X , the path components of $p^{-1}(A)$ are edges of E , and each is mapped by p homeomorphically onto A . Thus E has k times as many edges as X , and k times as many vertices. It follows that $\chi(E) = k\chi(X)$. Since the fundamental group of X has $n + 1$ free generators, the preceding lemma tells us that $\chi(X) = -n$. Then the number of free generators of the fundamental group of E , which is isomorphic to H , is

$$1 - \chi(E) = 1 - k\chi(X) = 1 + kn. \quad \blacksquare$$

Note that if F is a free group with a finite system of free generators and H is a subgroup of F such that F/H is infinite, then nothing can be said about the cardinality of a system of free generators for H . It might be finite (for instance, if H is the trivial subgroup) or infinite (for instance, if H is the fundamental group of the covering space pictured in Example 2 of §81).

Exercises

1. Show that the Euler number of a finite linear graph X is a topological invariant of X . [Hint: First consider the case where X is connected.]
2. Let F be a free group on two free generators α and β . Let H be the subgroup generated by α . Show that H has infinite index in F .
3. Let $p : \mathbb{R} \rightarrow S^1$ be the standard covering map; consider the covering map $p \times p : \mathbb{R} \times \mathbb{R} \rightarrow S^1 \times S^1$. Let $b_0 = (1, 0) \in S^1$; set $X = (b_0 \times S^1) \cup (S^1 \times b_0)$; let $E = (p \times p)^{-1}(X)$; and let $q : E \rightarrow X$ be the covering map obtained by restricting $p \times p$. The fundamental group of X has free generators α and β , where α is represented by a loop in $b_0 \times S^1$ and β by a loop in $S^1 \times b_0$. Find a system of free generators for the subgroup $q_*(\pi_1(E, e_0))$, where e_0 is the origin in $\mathbb{R}^2$.